



Probability and Statistics

ES-111 Spring 2025

Assignment 1; Chapter 2

Deadline of submission: 10.02.2025

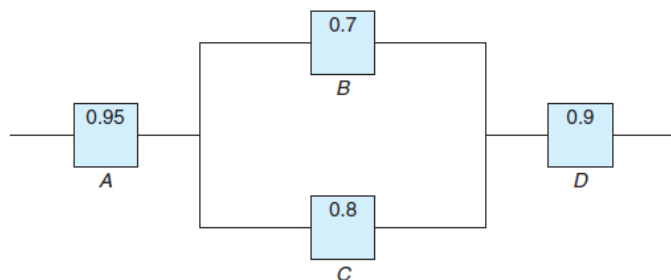
- Q1.** If a multiple-choice test consists of 5 questions each with 4 possible answers of which only one is correct, **(a)** How many different ways can a student check off one answer to each question? **(b)** How many ways can a student check off one answer to each question and get all the questions wrong?
- Q2.** A shipment of 12 television sets contains 3 defective sets. In how many ways can a hotel purchase 5 of these sets and receive at least 2 of the defective?
- Q3.** **(a)** How many three-digit numbers can be formed from the digits 0, 1, 2, 3, 4, 5, and 6, if each digit can be used only once? **(b)** How many of these are odd numbers? **(c)** How many are greater than 330?
- Q4.** Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated **(a)** With no restrictions? **(b)** If each couple is to sit together? **(c)** If all the men sit together to the right of all the women?
- Q5.** The probability that an American industry will locate in Shanghai, China is 0.7, the probability that it will locate in Beijing, China is 0.4, and the probability that it will locate in either Shanghai or Beijing or both is 0.8. What is the probability that the industry will locate **(a)** in both cities? **(b)** in neither city?
- Q6.** If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary, what is the probability that **(a)** the dictionary is selected? **(b)** 2 novels and 1 book of poems are selected?
- Q7.** The probability that a married man watches a certain television show is 0.4 and the probability that a married woman watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find the probability that **(a)** a married couple watches the show; **(b)** a wife watches the show given that her husband does; **(c)** at least 1 person of a married couple will watch the show.

Q8. The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient enters a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis, and the patient sues?

Q9. Before the distribution of certain statistical software every fourth compact disk (CD) is tested for accuracy. The testing process consists of running four independent programs and checking the results. The failure rate for the 4 testing programs are respectively, 0.01, 0.03, 0.02, and 0.01. **(a)** What is the probability that a CD was tested and failed any test? **(b)** Given that a CD was tested, what is the probability that it failed program 2 or 3? **(c)** In a sample of 100, how many CDs would you expect to both rejected. **(d)** Given a CD was defective, what is the probability that it was tested?

Q10. One overnight case contains 2 bottles of aspirin and 3 bottles of thyroid tablets. A second tote bag contains 3 bottles of aspirin, 2 bottles of thyroid tablets, and 1 bottle of laxative tablets. If 1 bottle of tablets is taken at random from each piece of luggage, find the probability that **(a)** both bottles contain thyroid tablets; **(b)** neither bottle contains thyroid tablets; **(c)** the 2 bottles contain different tablets.

Q11. Suppose the diagram of an electrical system is as given in following figure. What is the probability that the system works? Assume the components fail independently.



Q12. If the probability is 0.1 that a person will make a mistake on his or her state income tax return, find the probability that **(a)** four totally unrelated persons each make a mistake; **(b)** Mr. Jones and Ms. Clark both make a mistake, and Mr. Roberts and Ms. Williams do not make a mistake.

Q13. The probability that a patient recovers from a delicate heart operation is 0.8. What is the probability that **(a)** exactly 2 of the next 3 patients who have this operation survive? **(b)** all of the next 3 patients who have this operation survive.

Q15. Two jurors are selected from 4 alternates to serve at a murder trial. Using the notation A_1A_3 , for example, to denote the simple event that alternates 1 and 3 are selected, list the 6 elements of the sample space S .

Q16. Four students are selected at random from a chemistry class and classified as male or female. List the elements of the sample space S_1 , using the letter M for male and F for female.

Define a second sample space S_2 where the elements represent the number of females selected.

Q17. A developer of a new subdivision offers a prospective home buyer a choice of 4 designs, 3 different heating systems, a garage or carport, and a patio or screened porch. How many different plans are available to this buyer?

Q18. In a fuel economy study, each of 3 race cars is tested using 5 different brands of gasoline at 7 test sites located in different regions of the country. If 2 drivers are used in the study, and test runs are made once under each distinct set of conditions, how many test runs are needed?

Q19. In how many ways can 6 people be lined up to get on a bus?

(a) If 3 specific persons, among 6, insist on following each other, how many ways are possible?

(b) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?

Q20. (a) How many distinct permutations can be made from the letters of the word *COLUMNS*?

(b) How many of these permutations start with the letter *M*?

Q21. Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated (a) with no restrictions? (b) if each couple is to sit together?

Q22. How many ways are there that no two students will have the same birth date in a class of size 60?

Q23. In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that

(a) the student took mathematics or history; (b) the student did not take either of these subjects; (c) the student took history but not mathematics.

Q24. Interest centers around the life of an electronic component. Suppose it is known that the probability that the component survives for more than 6000 hours is 0.42. Suppose also that the probability that the component survives *no longer than* 4000 hours is 0.04.

(a) What is the probability that the life of the component is less than or equal to 6000 hours?

(b) What is the probability that the life is greater than 4000 hours?

Q25. A truth serum has the property that 90% of the guilty suspects are properly judged while, of course, 10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent?

Q26. From a box containing 6 black balls and 4 green balls, 3 balls are drawn in succession, each ball being replaced in the box before the next draw is made. What is the probability that (a) all 3 are the same color? (b) each color is represented?

Q27. A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?

Q28. A rare disease exists with which only 1 in 500 is affected. A test for the disease exists, but of course it is not infallible. A correct positive result (patient actually has the disease) occurs 95% of the time, while a false positive result (patient does not have the disease) occurs 1% of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?

Bonus Question - Sleeping beauty dilemma.

On Sunday, Sleeping Beauty agrees to participate in an interesting experiment, run by a professor. She gets a sleeping pill and goes to sleep. The professor then tosses a fair coin. From now on, there are two possibilities:

- the coin comes up heads. In this case, the professor will awake Sleeping Beauty on Monday and ask her: “What is the probability that the coin came up heads?” After this, the experiment ends.
- the coin comes up tails, in which case Sleeping Beauty is also awakened on Monday, asked the same question, but put to sleep again and awakened on Tuesday, to once again be asked the same question.

Note that the sleeping pill also causes Sleeping Beauty to forget any earlier awakening: her memory is the exact same whether she awakes on Monday or on Tuesday.

Sleeping Beauty awakens. She knows about the structure of the experiment, but naturally doesn't know whether it is Monday or Tuesday. The professor asks: "What is the probability that the coin came up heads?"

Try to answer by yourself. If you are interested to learn more, watch this YouTube video (https://www.youtube.com/watch?v=XeSu9fBJ2sl&ab_channel=Veritasium)